BUSINESS ANALYTICS · TASK 1

Retail Sales Analysis & Executive Dashboard

Data cleaning, business calculations, pivot analysis and an interactive Excel dashboard for an 8-branch retail chain — FY2025

₹6,805,042
net revenue

29.2%
gross margin

1,602 → 1,560
rows cleaned

10
cleaning steps

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PROGRAMME
Sumerix Global · Business Analytics · 36 weeks

BATCH START
15 October 2026

TOOLS
Excel (Tables, SUMIFS, XLOOKUP, Power Query, Power Pivot / DAX)

1 · Executive summary

₹6,805,042	29.2%	1,560	99%
FY2025 net revenue	gross margin	orders · AOV ₹4,362	of company target

FY2025 net revenue ₹6,805,042 from 1,560 orders (AOV ₹4,362); gross margin 29.2%; plan attainment 99%.

Business problem. A retail chain with 8 branches receives raw branch spreadsheets every month. Management cannot see trends, top products, customer behaviour or regional performance without days of manual work. This project cleans the data, builds business calculations and pivot views, and delivers an interactive Excel dashboard with an executive report.

Dataset. DECLARED SYNTHETIC — 1,602 raw order lines for FY2025 generated with a fixed seed to mimic 8 Indian branches, 30 products, 5 categories and 320 customers. Data-quality problems were injected on purpose so every cleaning step is real. No real company data is used.

2 · Data cleaning & validation (raw 1,602 → clean 1,560 rows)

Step	Issue found	Count	Excel feature / formula	Result
1	Exact duplicate rows	36	`Data ▸ Remove Duplicates (all columns)`	Rows deleted
2	Leading/trailing/double spaces in Branch, Category, Customer Name	225	`=TRIM() helper column, then Paste Values`	Cells standardised
3	Mixed case in Customer Name (UPPER / lower)	70	`=PROPER(TRIM(F2))`	Cells standardised
4	Inconsistent Category spellings (ELECTRONICS, electronics, padded)	110	`Find & Replace + =PROPER(TRIM()) checked against Category list`	Cells standardised
5	Blank Region	12	`=INDEX(Branch_Master[Region],MATCH(C2,Branch_Master[Branch],0))`	Cells filled from master
6	Blank Payment Mode	15	`Go To Special ▸ Blanks ▸ type 'Not Recorded' ▸ Ctrl+Enter`	Cells filled (flagged, not guessed)
7	Order Date stored as text (dd/mm/yyyy)	28	`Data ▸ Text to Columns ▸ Date: DMY`	Converted to real dates
8	Discount stored as text ('10%')	22	`=VALUE(SUBSTITUTE(P2,"%",""))/100`	Converted to numbers
9	Zero / negative Quantity (invalid sale lines)	6	`Filter Qty <= 0 ▸ delete rows; Data Validation added`	Rows deleted
10	Validation rules for future entry	4	`Data ▸ Data Validation: Qty whole >0, Discount 0–50%, Region & Payment lists`	Rules applied

Reconciliation is live in the workbook: raw rows − 36 exact duplicates − 6 invalid-quantity rows = 1,560 = rows in `tblSales` ✔.

3 · Business calculations

KPI	Value	Logic
Gross Sales	₹7,225,866	Σ Quantity × Unit Price
Discount Given	₹420,824	Σ Gross × Discount %
Net Revenue	₹6,805,042	Gross − Discount
Cost of Goods Sold	₹4,816,440	Σ Quantity × Unit Cost
Gross Profit	₹1,988,602	Net Revenue − COGS
Gross Margin %	29.2%	Gross Profit ÷ Net Revenue
Effective Discount %	5.8%	Discount ÷ Gross Sales
Orders	1,560	Count of order IDs
Units Sold	3,314	Σ Quantity
Average Order Value	₹4,362	Net Revenue ÷ Orders
Active Customers	312	Distinct Customer IDs
H1 Net Revenue (Jan–Jun)	₹3,152,995	SUMIFS Month No ≤ 6
H2 Net Revenue (Jul–Dec)	₹3,652,047	SUMIFS Month No > 6
H2 vs H1 Growth %	15.8%	(H2 − H1) ÷ H1
Q3 Net Revenue	₹1,385,107	None
Q4 Net Revenue	₹2,266,939	None
Q4 vs Q3 Growth %	63.7%	(Q4 − Q3) ÷ Q3
Company Target 2025	₹6,900,000	Σ branch targets
Target Achievement %	98.6%	Net Revenue ÷ Target

4 · Pivot analysis

Net revenue by region × category

Region	Electronics	Home & Kitchen	Fashion	Grocery	Beauty	Total	Share
North	₹745,444	₹336,039	₹307,961	₹141,298	₹138,435	₹1,669,176	24.5%
South	₹1,067,825	₹557,003	₹350,955	₹221,562	₹166,536	₹2,363,881	34.7%
West	₹718,882	₹439,171	₹331,459	₹134,923	₹155,258	₹1,779,693	26.2%
East	₹517,996	₹180,127	₹137,225	₹75,151	₹81,791	₹992,290	14.6%
Total	₹3,050,146	₹1,512,339	₹1,127,601	₹572,934	₹542,021	₹6,805,042	100.0%

Category profitability

Category	Net revenue	Gross profit	Margin	Units
Electronics	₹3,050,146	₹639,696	21.0%	571
Home & Kitchen	₹1,512,339	₹429,889	28.4%	505
Fashion	₹1,127,601	₹551,731	48.9%	773
Grocery	₹572,934	₹130,914	22.8%	1022
Beauty	₹542,021	₹236,371	43.6%	443

Top 10 products

#	Product	Category	Net revenue	Margin
1	Smartphone X1	Electronics	₹1,218,669	15.8%
2	32-inch Smart TV	Electronics	₹880,745	15.0%
3	RO Water Purifier	Home & Kitchen	₹456,454	22.2%
4	Air Fryer 4L	Home & Kitchen	₹377,506	24.6%
5	Smartwatch Fit	Electronics	₹321,733	30.7%
6	Running Shoes	Fashion	₹309,010	39.9%
7	Mixer Grinder 750W	Home & Kitchen	₹262,270	24.7%
8	Women's Kurta Set	Fashion	₹250,668	50.1%
9	Wi-Fi Router AX	Electronics	₹229,456	28.6%
10	Men's Denim Jeans	Fashion	₹228,177	50.4%

Branch performance vs target

Branch	Region	Net revenue	Target	Achievement	Status	Orders	AOV
Delhi	North	₹1,087,946	₹1,200,000	90.7%	Below target	257	₹4,233
Ghaziabad	North	₹581,230	₹500,000	116.2%	Achieved	137	₹4,243
Mumbai	West	₹1,032,503	₹1,000,000	103.3%	Achieved	249	₹4,147
Pune	West	₹747,190	₹700,000	106.7%	Achieved	158	₹4,729

Chennai	South	₹835,296	₹900,000	92.8%	Below target	181	₹4,615
Bengaluru	South	₹957,514	₹900,000	106.4%	Achieved	226	₹4,237
Coimbatore	South	₹571,071	₹600,000	95.2%	Near target	136	₹4,199
Kolkata	East	₹992,290	₹1,100,000	90.2%	Below target	216	₹4,594

5 · Interactive executive dashboard



Dashboard sheet (Region = All, Category = All, Channel = All). Changing the gold selector cells recalculates every KPI card, chart and table through SUMIFS with wildcard criteria.

Monthly trend

Month	Net revenue	Gross profit	Margin	MoM growth	Target	Achievement
Jan	₹633,573	₹172,203	27.2%	—	₹480,000	132.0%
Feb	₹424,357	₹136,907	32.3%	-33.0%	₹440,000	96.4%
Mar	₹561,879	₹161,209	28.7%	32.4%	₹510,000	110.2%
Apr	₹541,202	₹158,612	29.3%	-3.7%	₹480,000	112.8%
May	₹528,980	₹153,660	29.0%	-2.3%	₹510,000	103.7%
Jun	₹463,004	₹130,134	28.1%	-12.5%	₹470,000	98.5%
Jul	₹394,579	₹132,869	33.7%	-14.8%	₹500,000	78.9%
Aug	₹541,189	₹157,539	29.1%	37.2%	₹550,000	98.4%
Sep	₹449,340	₹154,320	34.3%	-17.0%	₹590,000	76.2%
Oct	₹779,703	₹204,253	26.2%	73.5%	₹770,000	101.3%
Nov	₹775,372	₹228,192	29.4%	-0.6%	₹840,000	92.3%
Dec	₹711,864	₹198,704	27.9%	-8.2%	₹750,000	94.9%

6 · Findings

1. **Seasonality.** Q4 revenue was 64% above Q3 — October–November festive demand drives the year; H2 grew 16% over H1.
2. **Mix.** Electronics is the largest category but also the thinnest-margin one (21.0%) vs Fashion (48.9%).
3. **Regions.** South region contributes 35% of revenue; 3 of 8 branches finished below target.
4. **Discounting.** Effective discount was 5.8% of gross sales (₹420,825) — concentrated in the festive months.

Business challenge. Reports were compiled manually from 8 branch files with duplicates, blank regions/payment modes, text dates and inconsistent category names — 10 cleaning steps were needed before analysis.

7 · Recommendations

1. Protect festive margin: cap Electronics discounts at 10% and shift promotion budget to Fashion bundles.
2. Branch reviews: use the Pivot_Summary target table monthly; branches below 95% get a stock-and-staffing review before Q4.
3. Fix data at source: the Data Validation rules on Clean_Data (quantity, discount, region, payment lists) should be copied into each branch template.
4. Next step (Task 2–3): load tblSales into Power Query / Power BI for automated monthly refresh instead of manual consolidation.

8 · Excel features used

Area	Features
Data preparation	Remove Duplicates, TRIM, PROPER, Find & Replace, Text to Columns, Go To Special, Data Validation lists
Lookups	XLOOKUP, INDEX/MATCH, VLOOKUP, HLOOKUP (monthly targets)
Aggregation	SUMIFS, COUNTIFS, AVERAGEIFS with wildcard criteria; SUMPRODUCT
Logic & text	IF, IFS, IFERROR, TEXT, TEXTJOIN, LARGE + INDEX/MATCH ranking
Presentation	Excel Tables, conditional formatting, KPI cards, line / column / doughnut / bar charts, PivotTable + Slicer steps (README sheet)